



Education

Shanghai Normal University Master of Engineering in Computer Science and Technology	Sept 2022 – June 2025
• Major Courses: Digital Image Processing (90), Digital Signal Processing (90), Computer Network and Simulation (94)	
Nanjing University of Aeronautics and Astronautics Jincheng College Bachelor of Engineering in Transportation	Sept 2017 – June 2021
• Major Courses: Linear Algebra (93), Advanced Mathematics (98), Probability Theory and Mathematical Statistics (100)	

Publications

• *Haoyu Zhai et al.* "Density-increment and cut-edge optimized clustering via minimum spanning forest" *Neurocomputing*(2026) (First author)

Proposed SMMSF, a novel clustering framework that constructs a minimum spanning forest to better capture both local and global structures. Introduced a density-increment peak strategy for adaptive splitting and a cut-edge-based distance for topology-aware merging. Designed the algorithm, implemented the framework, and carried out extensive evaluations on synthetic, UCI, and segmentation datasets. Demonstrated consistent superiority over 11 state-of-the-art methods in clustering accuracy, robustness to noise, and adaptability across diverse data distributions.

• *Haoyu Zhai et al.* "A Novel Outlier Detection Algorithm Based on Symmetry and Distance Ratio" *International Conference on Pattern Recognition*(2024) (First author)

Traditional distance- or density-based outlier detection often fails under varying densities and shapes. We proposed a novel algorithm that defines an outlier degree using symmetry ratio and distance ratio, with thresholds automatically set by the IQR method. I designed and implemented the algorithm, derived the metrics, and conducted experiments on 4 synthetic and 16 UCI datasets. Results showed strong accuracy and robustness, outperforming classical methods such as LOF, IForest, and AutoEncoder, while remaining efficient at $O(n\log n)$.

• *Zexuan Fei, Haoyu Zhai et al.* "Discovering generalized clusters with adaptive mixture density-based clustering" *Knowledge-based Systems* (2025) (Co-first author)

We introduced an adaptive mixture density-based clustering algorithm to automatically discover generalized clusters without prior parameter tuning. The method adaptively estimates mixture densities and integrates them into a flexible clustering framework. I was primarily responsible for experiment design and implementation, conducting extensive evaluations on synthetic and real-world datasets, including noise and ablation studies. Results demonstrated consistent improvements over state-of-the-art clustering methods in accuracy, robustness, and adaptability.

Research Program

National Natural Science Foundation of China (NSFC) Project National-level program, acted as algorithm designer.	Sept 2022 – June 2025
• Resulted in 10 related papers and 4 patents; 3 papers were closely tied to my contributions.	

Skills

Programming Languages: C/C++, Python, Matlab
Frameworks & Tools: PyTorch, OpenCV, Scikit-Learn, Git, Docker, Linux, Milvus, LaTeX
Multimodal & LLM Systems: LLM Agents, Agent Security, Docker Sandboxing, MCP (Model Context Protocol), RAG Pipelines, Vector Databases
Machine Learning: Machine Learning, Deep Learning, Transformer Models, Computer Vision, Clustering, Anomaly Detection

Awards

- Outstanding Graduate Award (Shanghai Normal University), 2025.
- Shanghai Normal University Scholarship, 2023 2024 2025.
- NUA College of Jincheng Scholarship, 2018 2019 2020 2021.

Experience

AI Algorithm Engineer , Qihoo 360 Technology Co.,Ltd – Beijing, China	July 2025 – present
• Built MAGA (Make Antivirus Great Agent), an LLM-based autonomous agent system for security analysis using ReAct-style tool orchestration, Docker sandbox isolation, and binary/PE/Office/PDF malware analysis tools. Designed batch multi-agent dispatch for large-scale triage and implemented a FastAPI WebUI for task submission, progress tracking, and result visualization.	
• Developed large-scale code analysis pipelines for similarity search and anomaly detection, using LLMs and vector databases (Milvus).	
• Built internal LLM-based single-agent system with tool orchestration and Docker sandbox execution for automated analysis tasks.	